

ME 547: Linear Systems

State-Space Canonical Forms

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Goal

The realization problem:

- ▶ existence and uniqueness: the same system can have infinite amount of state-space representations: e.g.

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases} \qquad \begin{cases} \dot{x} &= Ax + \frac{1}{2}Bu \\ y &= 2Cx \end{cases}$$

$$\left(\text{packed representation: } \Sigma_1 = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right], \Sigma_2 = \left[\begin{array}{c|c} A & \frac{B}{2} \\ \hline 2C & D \end{array} \right] \right)$$

- ▶ canonical realizations exist
- ▶ relationship between different realizations?
- ▶ example problem for this set of notes:

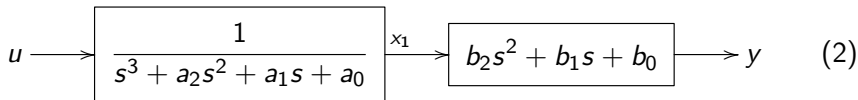
$$G(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0}. \quad (1)$$

Outline

1. CT controllable canonical form
2. CT observable canonical form
3. CT diagonal and Jordan canonical forms
4. Modified canonical form
5. DT state-space canonical forms
6. Similar realizations

Controllable canonical form (ccf)

Choose x_1 such that



$$X_1(s) = \frac{U(s)}{s^3 + a_2s^2 + a_1s + a_0} \Rightarrow \ddot{x}_1 + a_2\ddot{x}_1 + a_1\dot{x}_1 + a_0x_1 = u$$

Let $x_2 = \dot{x}_1$, $x_3 = \dot{x}_2$. Then $\dot{x}_3 = -a_2x_3 - a_1x_2 - a_0x_1 + u$.

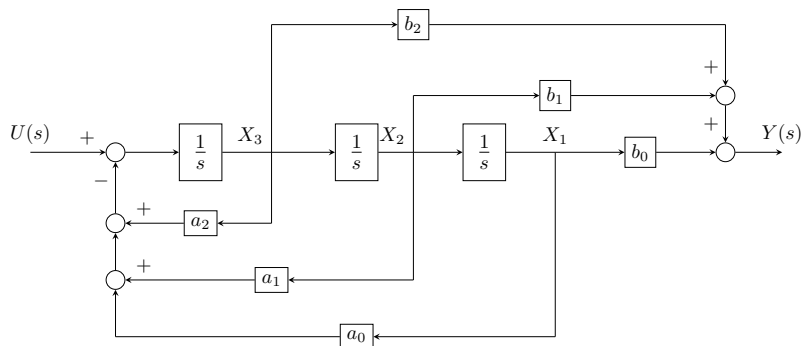
$$Y(s) = (b_2s^2 + b_1s + b_0) X_1(s) \Rightarrow y = b_2 \underbrace{\ddot{x}_1}_{x_3} + b_1 \underbrace{\dot{x}_1}_{x_2} + b_0x_1$$

Putting in matrix form yields

$$\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) \quad (3)$$
$$y(t) = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

Block diagram realization of controllable canonical forms

$$\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) \quad (4)$$
$$y(t) = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$



General ccf

For a single-input single-output transfer function

$$G(s) = \frac{b_{n-1}s^{n-1} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0} + d,$$

we can verify that

$$\Sigma_c = \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & D_c \end{array} \right] = \left[\begin{array}{ccccc|c} 0 & 1 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ \hline -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} & 1 \\ \hline b_0 & b_1 & \cdots & b_{n-2} & b_{n-1} & d \end{array} \right] \quad (5)$$

realizes $G(s)$. This realization is called the *controllable canonical form*.

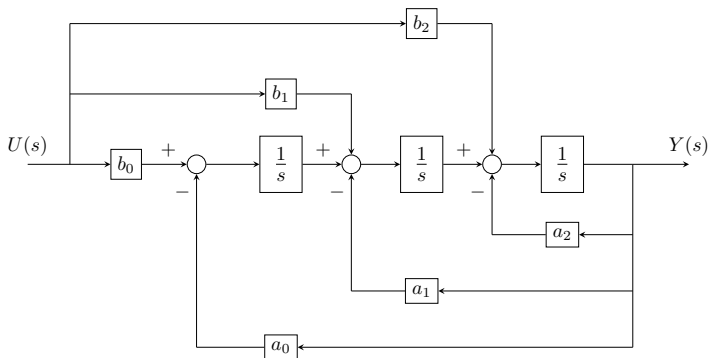
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Observable canonical form (ocf)

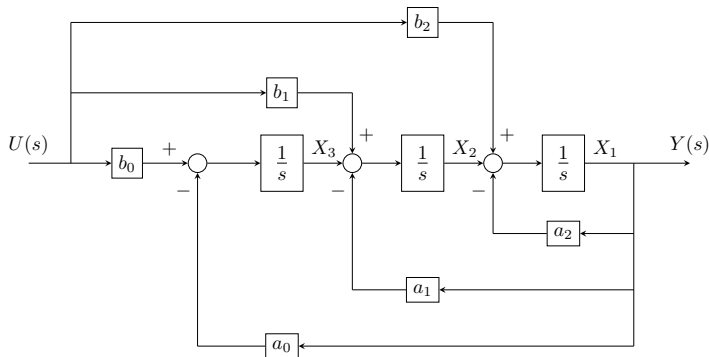
$$Y(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} U(s)$$

$$\Rightarrow Y(s) = -\frac{a_2}{s} Y(s) - \frac{a_1}{s^2} Y(s) - \frac{a_0}{s^3} Y(s) + \frac{b_2}{s} U(s) + \frac{b_1}{s^2} U(s) + \frac{b_0}{s^3} U(s).$$

In a block diagram, the above looks like



Observable canonical form



Here, the states are connected by

$$\begin{aligned}
 Y(s) &= X_1(s) & y(t) &= x_1(t) \\
 sX_1(s) &= -a_2X_1(s) + X_2(s) + b_2U(s) & \dot{x}_1(t) &= -a_2x_1(t) + x_2(t) + b_2u(t) \\
 sX_2(s) &= -a_1X_1(s) + X_3(s) + b_1U(s) & \dot{x}_2(t) &= -a_1x_1(t) + x_3(t) + b_1u(t) \\
 sX_3(s) &= -a_0X_1(s) + b_0U(s) & \dot{x}_3(t) &= -a_0x_1(t) + b_0u(t)
 \end{aligned}
 \Rightarrow$$

Observable canonical form

$$\begin{cases} \dot{x}_1(t) &= -a_2 x_1(t) + x_2(t) + b_2 u(t) \\ \dot{x}_2(t) &= -a_1 x_1(t) + x_3(t) + b_1 u(t) \\ \dot{x}_3(t) &= -a_0 x_1(t) + b_0 u(t) \\ y(t) &= x_1(t) \end{cases}$$
$$\Rightarrow \dot{x}(t) = \underbrace{\begin{bmatrix} -a_2 & 1 & 0 \\ -a_1 & 0 & 1 \\ -a_0 & 0 & 0 \end{bmatrix}}_{A_o} x(t) + \underbrace{\begin{bmatrix} b_2 \\ b_1 \\ b_0 \end{bmatrix}}_{B_o} u(t)$$
$$y(t) = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{C_o} x(t)$$

The above is called the *observable canonical form* realization of $G(s)$.

Exercise

Verify that $C_o(sI - A_o)^{-1}B_o = G(s)$.

General ocf

In the general case, the *observable canonical form* of the transfer function

$$G(s) = \frac{b_{n-1}s^{n-1} + \cdots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0} + d$$

is

$$\Sigma_o = \left[\begin{array}{c|c} A_o & B_o \\ \hline C_o & D_o \end{array} \right] = \left[\begin{array}{ccccc|c} -a_{n-1} & 1 & \cdots & 0 & 0 & b_{n-1} \\ -a_{n-2} & 0 & \cdots & 0 & 0 & b_{n-2} \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ -a_1 & 0 & \cdots & 0 & 1 & b_1 \\ -a_0 & 0 & \cdots & 0 & 0 & b_0 \\ \hline 1 & 0 & \cdots & 0 & 0 & d \end{array} \right]. \quad (6)$$

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Diagonal form

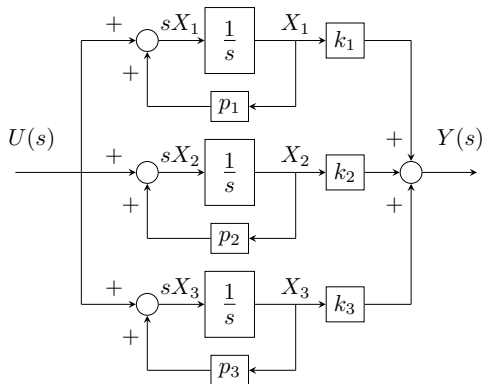
When

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0}$$

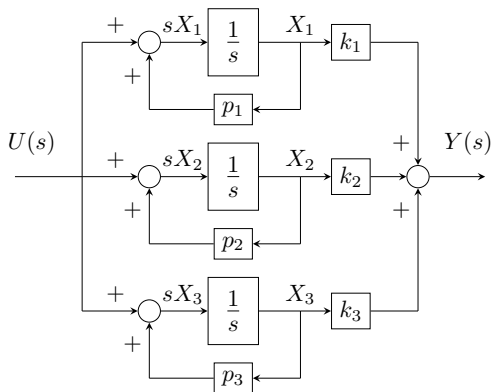
and the poles $p_1 \neq p_2 \neq p_3$, partial fractional expansion yields

$$G(s) = \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2} + \frac{k_3}{s - p_3}, \quad k_i = \lim_{p \rightarrow p_i} (s - p_i) \frac{B(s)}{A(s)},$$

namely



Diagonal form



The state-space realization of the above is

$$A = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_2 & 0 \\ 0 & 0 & p_3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad C = [k_1 \quad k_2 \quad k_3], \quad D = 0.$$

Jordan form

If poles repeat, say,

$$G(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} = \frac{b_2 s^2 + b_1 s + b_0}{(s - p_1)(s - p_m)^2}, \quad p_1 \neq p_m \in \mathbb{R},$$

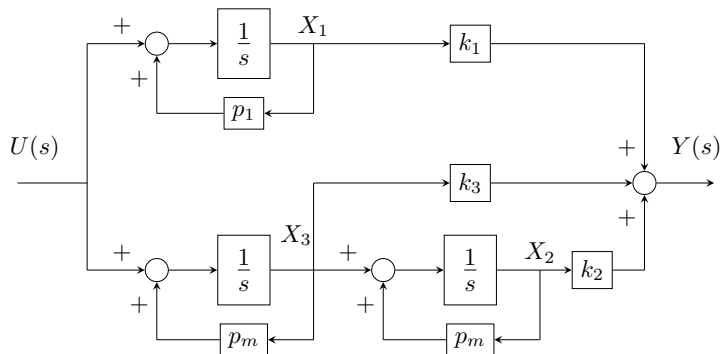
then partial fraction expansion gives

$$G(s) = \frac{k_1}{s - p_1} + \frac{k_2}{(s - p_m)^2} + \frac{k_3}{s - p_m} \quad \text{w/} \quad \begin{cases} k_1 &= \lim_{s \rightarrow p_1} G(s)(s - p_1) \\ k_2 &= \lim_{s \rightarrow p_m} G(s)(s - p_m)^2 \\ k_3 &= \lim_{s \rightarrow p_m} \frac{d}{ds} \{ G(s)(s - p_m)^2 \} \end{cases}$$

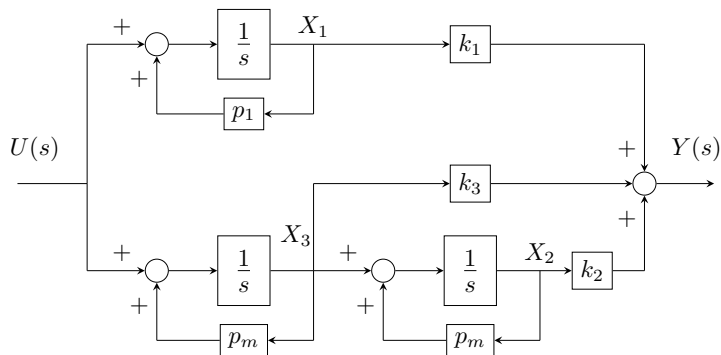
Jordan form

$$G(s) = \frac{k_1}{s - p_1} + \frac{k_2}{(s - p_m)^2} + \frac{k_3}{s - p_m}$$

has the block diagram realization:



Jordan form



The state-space realization of the above, called the Jordan canonical form, is

$$A = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_m & 1 \\ 0 & 0 & p_m \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad C = [k_1 \quad k_2 \quad k_3], \quad D = 0.$$

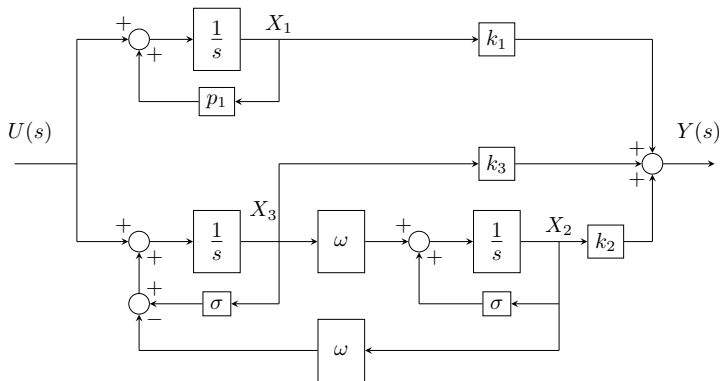
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Modified canonical form

If the system has complex poles, say,

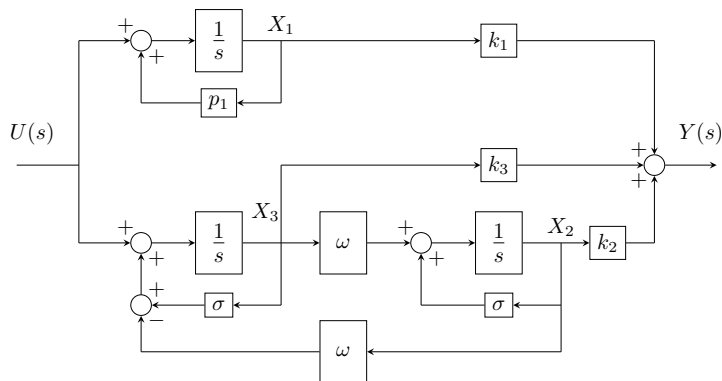
$$G(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} = \frac{k_1}{s - p_1} + \frac{\alpha s + \beta}{(s - \sigma)^2 + \omega^2}$$

then we have



where $k_2 = (\beta + \alpha\sigma)/\omega$ and $k_3 = \alpha$.

Modified canonical form



$\Rightarrow$ *modified Jordan form*:

$$A = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & \sigma & \omega \\ 0 & -\omega & \sigma \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad C = [k_1 \quad k_2 \quad k_3], \quad D = 0.$$

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DT state-space canonical forms

- ▶ The procedures for finding state space realizations in discrete time is similar to the continuous time cases. The only difference is that we use

$$\mathcal{Z} \{x(k+n)\} = z^n X(z),$$

instead of

$$\mathcal{L} \left\{ \frac{d^n}{dt^n} x(t) \right\} = s^n X(s),$$

assuming zero state initial conditions.

- ▶ Fundamental relationships:

$$x(k) \longrightarrow \boxed{z^{-1}} \longrightarrow x(k-1)$$

$$X(z) \longrightarrow \boxed{z^{-1}} \longrightarrow z^{-1}X(z)$$

$$x(k+n) \longrightarrow \boxed{z^{-1}} \longrightarrow x(k+n-1)$$

$$G(z) = \frac{b_2 z^2 + b_1 z + b_0}{z^3 + a_2 z^2 + a_1 z + a_0}$$

- ▶ Same transfer-function structure $\Rightarrow$ same A , B , C , D matrices of the canonical forms as those in continuous-time cases
- ▶ Controllable canonical form:

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

$$G(z) = \frac{b_2 z^2 + b_1 z + b_0}{z^3 + a_2 z^2 + a_1 z + a_0}$$

► Observable canonical form:

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} -a_2 & 1 & 0 \\ -a_1 & 0 & 1 \\ -a_0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} b_2 \\ b_1 \\ b_0 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

DT diagonal form

$$G(z) = \frac{b_2 z^2 + b_1 z + b_0}{z^3 + a_2 z^2 + a_1 z + a_0}$$

► Diagonal form (distinct poles):

$$G(z) = \frac{k_1}{z - p_1} + \frac{k_2}{z - p_2} + \frac{k_3}{z - p_3}$$

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_2 & 0 \\ 0 & 0 & p_3 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(k)$$
$$y(k) = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

DT Jordan form 1

$$G(z) = \frac{b_2 z^2 + b_1 z + b_0}{z^3 + a_2 z^2 + a_1 z + a_0}$$

► Jordan form (2 repeated poles):

$$G(z) = \frac{k_1}{z - p_1} + \frac{k_2}{(z - p_m)^2} + \frac{k_3}{z - p_m}$$

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_m & 1 \\ 0 & 0 & p_m \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

DT Jordan form 2

$$G(z) = \frac{b_2 z^2 + b_1 z + b_0}{z^3 + a_2 z^2 + a_1 z + a_0}$$

► Jordan form (2 complex poles):

$$G(s) = \frac{k_1}{z - p_1} + \frac{\alpha z + \beta}{(z - \sigma)^2 + \omega^2}$$

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & \sigma & \omega \\ 0 & -\omega & \sigma \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u(k)$$
$$y(k) = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

where $k_2 = (\beta + \alpha\sigma)/\omega$, $k_3 = \alpha$.

Exercise

obtain the controllable canonical form:

$$\blacktriangleright G(s) = \frac{z^{-1} - z^{-3}}{1 + 2z^{-1} + z^{-2}}$$

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Relation between different realizations

Similar realizations

Given one realization Σ of a transfer function $G(s)$ and a nonsingular $T \in \mathbb{R}^{n \times n}$, we can define *new* states:

$$Tx^* = x.$$

Then

$$\dot{x}(t) = Ax(t) + Bu(t) \Rightarrow \frac{d}{dt} (Tx^*(t)) = ATx^*(t) + Bu(t),$$

$$\Rightarrow \Sigma^* : \begin{cases} \dot{x}^*(t) &= T^{-1}ATx^*(t) + T^{-1}Bu(t) \\ y(t) &= CTx^*(t) + Du(t) \end{cases}$$

namely

$$\Sigma^* = \left[\begin{array}{c|c} T^{-1}AT & T^{-1}B \\ \hline CT & D \end{array} \right]$$

also realizes $G(s)$ and is said to be *similar* to Σ .

Relation between different realizations

Similar realizations

Exercise (Another observable canonical form.)

Verify that the following realize the same system

$$\Sigma = \left[\begin{array}{ccc|c} -a_2 & 1 & 0 & b_2 \\ -a_1 & 0 & 1 & b_1 \\ -a_0 & 0 & 0 & b_0 \\ \hline 1 & 0 & 0 & d \end{array} \right], \quad \Sigma^* = \left[\begin{array}{ccc|c} 0 & 0 & -a_0 & b_0 \\ 1 & 0 & -a_1 & b_1 \\ 0 & 1 & -a_2 & b_2 \\ \hline 0 & 0 & 1 & d \end{array} \right]$$